

PDC Reconstruction Literature Survey and SAMURAI Terminal Air-MS Gap Analysis

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1 TL;DR

This document surveys the published reference frame for the SAMURAI PDC1/PDC2 drift-chamber chain and contrasts it with our current Geant4 reconstruction performance, in order to localise the air multiple-scattering gap that drives our PDC2 spatial resolution one order of magnitude above design.

- **Reference frame.** The PDC1/PDC2 chain has no dedicated peer-reviewed publication. The canonical references are the SAMURAI overview paper by Kobayashi et al., Nucl. Instr. Meth. B **317**, 294–304 (2013) §3.4¹, the RIKEN internal technical note *Detector-PDC.pdf*², and the Kahlbow PhD thesis (TU Darmstadt, 2019)³ which documents an independent reconstruction of the same hardware.
- **Design specifications.** Combining Kobayashi 2013 §3.4¹ with the SAMURAI 2012 construction proposal⁴, the PDC1/PDC2 chain targets 1 mm position resolution, 2 mrad angular resolution, and 0.8% relative momentum resolution ($\Delta p/p$). Commissioning RI-beam runs reached $\Delta p/p \approx 1/1500$ ¹; the original proton-mode design value is $\Delta p/p \approx 1/700$ as quoted in the RIBF-TAC05 (2005) construction proposal⁵.
- **Our measured σ_y (PDC2).** In our Geant4 + reconstruction chain we observe σ_y (PDC2) = 12.83–14.93 mm in the all-air configuration, 6.80–8.21 mm when Region A (dipole cavity + beam pipe) is set to vacuum (`beamline_vacuum`), and approximately 1.0 mm in the all-vacuum configuration⁶. The realistic-air values sit roughly an order of magnitude away from the 1 mm hardware design specification¹.
- **Air MS dominates and the configuration choice matters.** A quadrature decomposition⁶ attributes the all-air budget to $\sigma_A \approx 11.19$ mm from Region A (cavity + beam pipe, ~ 4.3 m) and $\sigma_B \approx 7.43$ mm from Region B (Exit Window $\rightarrow$ PDCs, ~ 2 m). The stage D inputs from the same memo⁷ give σ_y (PDC2) = 7.49 mm if Region A is vacuum and σ_y (PDC2) = 13.50 mm if both Region A and Region B are air. Whether Region A is air or vacuum during the experiment is therefore a configuration question that selects between these two values; this document does not assert which one corresponds to the actual run.

See §6 for proposed action items.

2 The problem: air multiple scattering at the SAMURAI terminal

2.1 Geometry of the proton flight path

The deuteron target sits at world coordinates $(-12.488, 0.0009, -1069.458)$ mm, which corresponds to yoke-local coordinates $(-545.5, 0, -919.9)$ mm under the -30° yoke rotation about Y ⁸. Because the SAMURAI dipole cavity (`vchamber_box`) extends over $z \in [-1750, +1750]$ mm

¹T. Kobayashi et al., Nucl. Instr. Meth. B **317**, 294–304 (2013), DOI: [10.1016/j.nimb.2013.05.089](https://doi.org/10.1016/j.nimb.2013.05.089).

²RIKEN Nishina Center, *Detector-PDC* technical note, <https://www.nishina.riken.jp/ribf/SAMURAI/image/Detector-PDC.pdf>.

³J. Kahlbow, PhD thesis, TU Darmstadt (2019), TUpriints 9192, <https://tuprints.ulb.tu-darmstadt.de/9192/>.

⁴SAMURAI Collaboration, *SAMURAI Construction Proposal* (2012), https://ribf.riken.jp/SAMURAI/120425_SAMURAIConstProp.pdf.

⁵SAMURAI Collaboration, *RIBF-TAC05 SAMURAI construction proposal* (2005), https://ribf.riken.jp/RIBF-TAC05/10_SAMURAI.pdf.

⁶See `docs/reports/reconstruction/momentum_reco/ms_ablation/ms_ablation_mixed_20260422_zh.md` §4.

⁷See `docs/reports/reconstruction/momentum_reco/ms_ablation/ms_ablation_mixed_20260422_zh.md` §6.

⁸See `docs/reports/reconstruction/momentum_reco/ms_ablation/ms_ablation_mixed_20260422_zh.md` §1.1–§1.3 for coordinates, yoke rotation, and Region A/B definitions.

in yoke-local frame, the target at $z = -919.9$ mm is *physically inside the dipole cavity* rather than upstream of it. Two regions therefore separate the target from the PDCs. **Region A** (≈ 4.3 m) covers the in-cavity flight from the target through the cavity exit (2.67 m) and onwards through the downstream beam-pipe (`VacuumDownstream`, ≈ 1.6 m) up to the Exit Window; this region is controlled by the Geant4 macro switch `/samurai/geometry/BeamLineVacuum`. **Region B** (≈ 2.0 m) covers free flight in the world volume from the Exit Window at world (170.5, 0, 3187.5) mm to PDC1 at (700, 0, 4000) mm (0.97 m) and on to PDC2 (1.00 m), gated by `/samurai/geometry/FillAir`. Inside Region A the proton bends $\sim 30^\circ$ in the 1.15 T field, but the arc-length vs. straight-line difference stays below 10%⁸.

2.2 Measured σ values

Table 1 reproduces the robust half-width $\sigma = (p_{84} - p_{16})/2$ measured at PDC1 and PDC2 across three vacuum/air configurations and three truth momenta, with $|\vec{p}| = 627$ MeV/c protons.

Condition	(p_x, p_y) [MeV/c]	N	$\sigma_x(\text{P1})$	$\sigma_y(\text{P1})$	$\sigma_x(\text{P2})$	$\sigma_y(\text{P2})$	σ_{θ_x}	σ_{θ_y}
all_air	(0, 0)	50	3.071	9.736	3.669	12.826	12.279	9.240
all_air	(100, 0)	50	3.838	10.492	5.377	12.751	15.768	8.277
all_air	(0, 20)	50	3.372	13.039	4.065	14.927	21.709	7.846
beamline_vacuum	(0, 0)	50	1.528	4.061	2.086	6.801	11.834	5.910
beamline_vacuum	(100, 0)	50	2.067	4.888	3.060	8.211	8.100	5.786
beamline_vacuum	(0, 20)	49	1.667	4.789	2.363	7.459	14.289	5.421
all_vacuum	(0, 0)	50	0.147	0.515	0.447	0.947	6.798	2.558
all_vacuum	(100, 0)	50	0.154	0.364	0.416	0.999	4.075	1.956
all_vacuum	(0, 20)	50	0.174	0.411	0.396	1.002	6.913	2.299

Table 1: σ values from Geant4 + reconstruction ensemble; σ in mm, σ_θ in mrad. Robust half-width $(p_{84} - p_{16})/2$. Source: docs/reports/reconstruction/momentum_reco/ms_ablation/ms_ablation_mixed_20260422_zh.md §4.

2.3 Gap to design specifications

The realistic-air spatial $\sigma_y(\text{PDC2})$ sits roughly an order of magnitude away from the 1 mm hardware design figure, and only the all-vacuum ablation reaches the design floor. Table 2 summarises the comparison; angular and momentum entries follow the same structure.

Quantity	Design / commissioning	Our current state	Source
Position σ at PDC	1 mm	~ 13 mm (all-air) / ~ 7.5 mm (A=vac) / ~ 1.0 mm (all-vac)	Kobayashi 2013 §3.4 ¹ + memo §4 ⁶
Angular σ	2 mrad	5.4–9.2 mrad (σ_{θ_y} , all-air & <code>beamline_vacuum</code>) / 2.0–2.6 mrad (all-vacuum)	Kobayashi 2013 §3.4 ¹ + memo §4 ⁶
RI commissioning $\Delta p/p$	1/1500	not measured in this ablation	Kobayashi 2013 ¹
Proton-mode design $\Delta p/p$	$\sim 1/700$	not measured in this ablation	RIBF-TAC05 (2005) ⁵

Table 2: PDC reconstruction performance: design / commissioning targets vs. our current Geant4 + reconstruction state. Position and angular rows compare against the Kobayashi 2013 §3.4 hardware specification; the two $\Delta p/p$ rows are not yet propagated end-to-end in our ablation, so we have $\sigma_y(\text{PDC2})$ but no single $\Delta p/p$ figure to report.

2.4 Why this matters for the science floor

The PDC2 σ_y entries above are not the end of the story: they propagate backwards through the Runge–Kutta tracker and the SAMURAI magnetic field map to a target-momentum uncertainty $\sigma_{p,\text{target}}$, which is the reconstruction science floor for any experiment in which the target sits inside the magnet and the outgoing proton is measured downstream (target $\rightarrow$ magnet $\rightarrow$ PDC). This document does not re-derive $\sigma_{p,\text{target}}$ propagation values; the per-method numerical breakdown lives in the RK error-analysis deep-dive memo⁹.

⁹See `docs/reports/reconstruction/momentum_reco/rk/rk_error_analysis_deep_dive_20260426_zh.pdf` for $\sigma_{p,\text{target}}$ propagation under MC, Profile, Laplace, MCMC, and MC-PDC.

3 PDC reconstruction reference frame

3.1 Hardware foundations

3.2 Algorithm spine

3.3 Performance achieved across SAMURAI campaigns

4 Gap analysis

4.1 Performance: σ values

4.2 Methodology

4.3 Material budget and vacuum configuration

4.4 Experimental configuration and geometry

5 Broader MINOS+SEASTAR chain context

5.1 SEASTAR programme timeline

5.2 Companion detectors

5.3 Reviews and resources

5.4 Caveats from the broader bibliography

6 Implications and next steps

6.1 Short-term data acquisition action items

6.2 Medium-term simulation and analysis directions

6.3 Open questions

6.4 Non-goals

7 References

7.1 Hardware and methodology (peer-reviewed, foundational)

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